

Data Model Documentation

Overview

The application manages the procurement and inventory lifecycle for a retail organization operating across India, with 150 stores, 12 warehouses, 2,000 suppliers, and 10,000 employees. The system handles the complete process from identifying a purchasing need through supplier selection, purchase approval, delivery, inventory updates, invoice verification, and payment reconciliation.

The system manages the following core domains:

- Procurement: Manages purchase requisitions, approvals, and purchase orders.
- Inventory: Tracks inventory levels, stock transfers, and goods receipts.
- Budgets: Monitors budget allocations, reservations, and spending.
- Suppliers: Manages supplier onboarding, quotations, and invoices.
- Payments: Handles payment processing and reconciliation.

Entity Relationship Diagram

```

[Users] ---- N:1 ---- [PurchaseRequisitions]
|
| 1:N
|
[Stores] ---- N:1 ---- [PurchaseRequisitions]
|
| 1:N
|
[CostCenters] ---- N:1 ---- [PurchaseRequisitions]
|
| 1:N
|
[FinancialYears] ---- N:1 ---- [PurchaseRequisitions]
|
| 1:N
|
[Products] ---- N:1 ---- [PurchaseRequisitionLines]
|
| 1:N
|
[PurchaseRequisitions] ---- 1:N ---- [PurchaseOrders]
|
| 1:N
|
[PurchaseOrders] ---- 1:N ---- [GoodsReceipts]
|
| 1:N
|
[GoodsReceipts] ---- 1:N ---- [Inventory]
|
| 1:N
|
[PurchaseOrders] ---- 1:N ---- [Invoices]
|
| 1:N
|
[Invoices] ---- 1:N ---- [Payments]

```

Entities

Users

Purpose: Represents users with different roles in the procurement and inventory management process.

Table/Collection Name: Users

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier

Field Name	Type	Required	Constraints	Description
username	String	Yes	Max 255 chars, unique	Username of the user
password	String	Yes	Min 8 chars	Password of the user
role	String	Yes	Must be one of predefined roles	Role of the user
created_at	Timestamp	Yes	Auto-set on create	Record creation time
updated_at	Timestamp	Yes	Auto-set on create/update	Last modification time

Relationships:

- Has many PurchaseRequisitions (1:N): Inverse via PurchaseRequisitions.user_id

Validation Rules:

- username must be non-empty and unique across all records
- password must be at least 8 characters long

PurchaseRequisitions

Purpose: Represents a request for purchasing goods or services.

Table/Collection Name: PurchaseRequisitions

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
user_id	UUID	Yes	Foreign key to Users	User who created the requisition
store_id	UUID	Yes	Foreign key to Stores	Store requesting the goods
cost_center_id	UUID	Yes	Foreign key to CostCenters	Cost center for the requisition
financial_year_id	UUID	Yes	Foreign key to FinancialYears	Financial year for the requisition

Field Name	Type	Required	Constraints	Description
status	String	Yes	Must be one of predefined statuses	Current status of the requisition
total_value	DECIMAL	Yes	Must be a positive number	Total value of the requisition
created_at	Timestamp	Yes	Auto-set on create	Record creation time
updated_at	Timestamp	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to Users (1:N): Foreign key user_id references Users.id
- Belongs to Stores (1:N): Foreign key store_id references Stores.id
- Belongs to CostCenters (1:N): Foreign key cost_center_id references CostCenters.id
- Belongs to FinancialYears (1:N): Foreign key financial_year_id references FinancialYears.id
- Has many PurchaseRequisitionLines (1:N): Inverse via PurchaseRequisitionLines.purchase_requisition_id

Validation Rules:

- total_value must be a positive number

PurchaseRequisitionLines

Purpose: Represents a line item within a purchase requisition.

Table/Collection Name: PurchaseRequisitionLines

Fields:

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
purchase_requisition_id	UUID	Yes	Foreign key to PurchaseRequisitions	Requisition this line belongs to
product_id	UUID	Yes	Foreign key to Products	Product being requested

Field Name	Type	Required	Constraints	Description
quantity	DECIMAL	Yes	Must be a positive number	Quantity of the product
estimated_price	DECIMAL	Yes	Must be a positive number	Estimated price of the product
required_delivery_date	DATE	Yes	Must be a valid date	Required delivery date
delivery_location_id	UUID	Yes	Foreign key to Stores or Warehouses	Location for delivery
cost_center_id	UUID	Yes	Foreign key to CostCenters	Cost center for the line
business_justification	String	Yes	Min 5 chars, Max 200 chars	Justification for the request
attachment	String	No	Max 255 chars	Attachment for the line
created_at	Timestamp	Yes	Auto-set on create	Record creation time
updated_at	Timestamp	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to PurchaseRequisitions (1:N): Foreign key purchase_requisition_id references PurchaseRequisitions.id
- Belongs to Products (1:N): Foreign key product_id references Products.id
- Belongs to Stores or Warehouses (1:N): Foreign key delivery_location_id references Stores.id or Warehouses.id
- Belongs to CostCenters (1:N): Foreign key cost_center_id references CostCenters.id

Validation Rules:

- quantity and estimated_price must be positive numbers
- required_delivery_date must be a valid date

Products

Purpose: Represents a product that can be purchased.

Table/Collection Name: Products**Fields:**

Field Name	Type	Required	Constraints	Description
id	UUID	Yes	Primary key, auto-generated	Unique identifier
sku	String	Yes	Max 255 chars, unique	SKU of the product
description	String	No	Max 255 chars	Description of the product
category_id	UUID	Yes	Foreign key to ProductCategories	Category of the product
unit	String	Yes	Max 255 chars	Unit of measure for the product
reorder_level	DECIMAL	Yes	Must be a positive number	Reorder level for the product
preferred_supplier_id	UUID	No	Foreign key to Suppliers	Preferred supplier for the product
batch_tracking_required	BOOLEAN	No	Default false	Indicates if batch tracking is required
expiry_tracking_required	BOOLEAN	No	Default false	Indicates if expiry tracking is required
created_at	Timestamp	Yes	Auto-set on create	Record creation time
updated_at	Timestamp	Yes	Auto-set on create/update	Last modification time

Relationships:

- Belongs to ProductCategories (1:N): Foreign key category_id references ProductCategories.id

- Optionally belongs to Suppliers (1:N): Foreign key preferred_supplier_id references Suppliers.id

Validation Rules:

- sku must be non-empty and unique across all records
- reorder_level must be a positive number

Relationships and Constraints

Foreign Keys

From Entity	Field	To Entity	Field	Cardinality	Delete Behavior
Users	id	PurchaseRequisitions	user_id	1:N	CASCADE
Stores	id	PurchaseRequisitions	store_id	1:N	CASCADE
CostCenters	id	PurchaseRequisitions	cost_center_id	1:N	CASCADE
FinancialYears	id	PurchaseRequisitions	financial_year_id	1:N	CASCADE
PurchaseRequisitions	id	PurchaseRequisitionLines	purchase_requisition_id	1:N	CASCADE
Products	id	PurchaseRequisitionLines	product_id	1:N	CASCADE
Stores	id	PurchaseRequisitionLines	delivery_location_id	1:N	CASCADE
CostCenters	id	PurchaseRequisitionLines	cost_center_id	1:N	CASCADE
PurchaseRequisitions	id	PurchaseOrders	purchase_requisition_id	1:N	CASCADE
PurchaseOrders	id	GoodsReceipts	purchase_order_id	1:N	CASCADE
PurchaseOrders	id	Invoices	purchase_order_id	1:N	CASCADE
Invoices	id	Payments	invoice_id	1:N	CASCADE

Unique Constraints

- `Users(username)`: Ensures no duplicate usernames
- `Products(sku)`: Ensures no duplicate SKUs

Data Validation Rules

Business Rules:

- `PurchaseRequisitions.total_value`: Must be a positive number, otherwise raise validation error "Total value must be positive".
- `PurchaseRequisitionLines.quantity`: Must be a positive number, otherwise raise validation error "Quantity must be positive".
- `PurchaseRequisitionLines.estimated_price`: Must be a positive number, otherwise raise validation error "Estimated price must be positive".

Format Rules:

- `Users.username`: Must be alphanumeric and between 2 and 50 characters, otherwise raise validation error "Invalid username format".
- `Products.sku`: Must be alphanumeric and between 1 and 255 characters, otherwise raise validation error "Invalid SKU format".

Data Lifecycle

Creation

- A new User is created when a new account is registered.
- Default status for `PurchaseRequisitions` is "DRAFT".
- `created_by` is automatically set to the current user for all entities.

Updates

- `User.password` can only be updated by the user themselves.
- `updated_at` is automatically updated on any modification to a record.

Archival / Soft Delete

- Records are not archived but can be deactivated by setting a `deleted_at` timestamp.

Indexes

- Primary key: `id`

- Foreign keys: user_id, store_id, cost_center_id, financial_year_id, purchase_requisition_id, product_id, delivery_location_id, purchase_order_id, invoice_id
- Search fields: username, sku

Example Records

Users

```
{
  "id": "550e8400-e29b-41d4-a716-446655440000",
  "username": "johndoe",
  "password": "encrypted_password",
  "role": "STORE_REQUESTER",
  "created_at": "2026-01-15T10:30:00Z",
  "updated_at": "2026-01-15T10:30:00Z"
}
```

PurchaseRequisitions

```
{
  "id": "123e4"
```